Innovation for Sustainable Development Goals

Water Scarcity Vulnerability Analysis

in Sfax Region, Tunisia

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Introduction

Water is among the most fundamental prerequisites for human life, agricultural productivity, and economic development. Yet across the globe, more than two billion people still lack access to safely managed drinking water, and the problem is intensifying under the compounding pressures of population growth, land-use change, and climate variability. The Middle East and North Africa (MENA) region already the most water-scarce on Earth faces a particularly acute crisis, with per capita renewable freshwater availability far below the global average and declining year on year.

Tunisia is no exception. While the country has made significant strides in expanding urban water infrastructure since independence, deep spatial inequalities persist between coastal urban centres and rural interior communities. The southern and central governorates endure chronic water stress from both sides: insufficient rainfall and falling groundwater tables reduce supply, while saltwater intrusion, industrial pollution, and ageing infrastructure degrade the quality of what little water remains accessible. Against this backdrop, Tunisia's commitments under the United Nations Sustainable Development Goals particularly SDG 6 (Clean Water and Sanitation) and SDG 13 (Climate Action) demand evidence-based, spatially explicit planning tools capable of guiding investment where it is most urgently needed.

This project focuses on the **Sfax Governorate**, Tunisia's largest governorate by area and its second most populous urban centre. Sfax presents a compelling case study precisely because it embodies the full complexity of water insecurity in a semi-arid Mediterranean context: coastal aquifer degradation from seawater intrusion and industrial effluents on one side; quantitative inland scarcity driven by erratic rainfall, high evapotranspiration, and over-extraction on the other. Between these two extremes, rural and peri-urban communities bear a disproportionate burden, relying on standpipes, water tankers, or untreated groundwater while the urban core enjoys comparatively reliable piped supply.

The response proposed in this study is an integrated **Water Scarcity Vulnerability Analysis** that brings together geospatial engineering, remote sensing, and socioeconomic data science to produce a multi-dimensional picture of water insecurity across the governorate. Drawing on satellite imagery processed in Google Earth Engine, hydrological models, open geospatial data from OpenStreetMap and HydroSHEDS, and national statistics from Tunisia's Institut National de la Statistique (INS) and SONEDE, the project constructs a **Climate–Water Vulnerability Index (CWVI)** that integrates climate stress, groundwater vulnerability, water quality, and access infrastructure into a single, policy-actionable map.

1. Study Area Definition

1.1. Location and Geographic Context

Study Area at a Glance

Region: Sfax Governorate, Tunisia
Position: Central-eastern coast, ≈ 3444′N, 1045′E
Area: ≈ 7,545 km^2 (largest governorate by area)
Population: ≈ 955,000 inhabitants
Climate: Semi-arid Mediterranean, with mean annual rainfall < 200 mm

Sfax Governorate is situated in the centre of Tunisia's eastern coastline, bordering the Gulf of Gabes. The region spans a coastal strip to the east and an arid interior plain to the west, creating a pronounced gradient in both climate and land use. The coastal zone hosts the city of Sfax, Tunisia's second largest urban centre and a major industrial and economic hub, while the interior is characterised by olive and cereal farming communities that depend almost entirely on groundwater for irrigation and domestic use.

1.2. Hydro-Geographic Structure

The region is divided into two distinct hydrological environments:

- **Coastal Zone:** Proximity to the Gulf of Gabes creates exposure to seawater intrusion into shallow coastal aquifers. Industrial activity and urban development amplify water quality degradation through pollution and over-extraction.
- **Inland Zone (Sfax–Agareb Basin):** Characterised by quantitative scarcity driven by low and erratic rainfall, limited surface water, and high evapotranspiration. Studies confirm that over 82% of groundwater samples in this basin fall into "poor" or "very poor" categories for drinking water quality.

1.3. Why This Study Area Was Chosen

The Sfax region was selected as the study area for a convergence of environmental, social, and policy-relevant reasons:

1. **Critical water stress:** Sfax combines both qualitative contamination (coastal) and quantitative scarcity (inland), making it an ideal test case for multi-dimensional vulnerability analysis.
2. **Agricultural dependency:** A significant share of the rural population relies on groundwater-fed olive orchards and cereal fields. Water insecurity translates directly into food insecurity and livelihood loss.

3. **Urban–rural disparities:** Urban areas enjoy near-universal pipe network coverage, while rural communities depend on public standpipes, tankers, or unsafe sources — a spatial inequality amenable to geospatial analysis.
4. **Climate exposure:** As a semi-arid region at the frontline of North African desertification, Sfax is highly exposed to climate-induced drought, rising temperatures, and shifting precipitation patterns.
5. **Policy relevance:** Tunisia's national water strategy and international SDG commitments create a direct demand for spatially explicit, evidence-based planning tools of the type this project delivers.

2. Research Objectives

The overarching aim of this project is to map and analyse access to water in the Sfax region in order to identify underserved and vulnerable areas, and to translate that spatial intelligence into actionable policy recommendations aligned with the SDGs.

Primary Objective

Develop a **Water Scarcity Vulnerability Map** for the Sfax region that integrates environmental, hydrological, climatic, and socioeconomic dimensions to identify priority zones for targeted intervention.

2.1. Specific Research Objectives

O1. Map spatial water access inequalities — Visualise the distribution of water infrastructure, network coverage, and service gaps, disaggregated by urban, peri-urban, and rural zones.

O2. Assess water quality risks — Analyse and map key quality indicators including salinity, contamination levels, and suitability for drinking and irrigation.

O3. Model groundwater vulnerability — Apply the DRASTIC model to evaluate aquifer susceptibility to contamination across the governorate.

O4. Quantify climate stress indicators — Calculate and map drought indices (SPI, SPEI, SMA), Land Surface Temperature (LST), and vegetation stress (NDVI) using satellite imagery processed in Google Earth Engine.

O5. Develop a composite vulnerability index — Combine the Climate–Water Vulnerability Index (CWVI = CSI × WSI) with agricultural exposure to produce spatially explicit risk hotspot maps.

O6. Propose spatial decision-support solutions — Design an interactive dashboard that translates vulnerability maps into investment scenarios quantifying impacts on water access, agricultural stability, and food security (SDG 2, 6, 13).

3. Data Collection

3.1. Spatial Data

Dataset	Description	Source
OpenStreetMap	Base map: water infrastructure, roads, settlements, administrative boundaries	Overpass Turbo
HydroSHEDS	River networks, watersheds, terrain hydrological layers	hydrosheds.org
Geofabrik OSM Extract	Shapefile export for Tunisia	geofabrik.de
Tunisia SDG ArcGIS Hub	Administrative layers and SDG indicators	arcgis.com
AQUASTAT	National water resource and irrigation datasets	FAO

Table 1: Spatial data sources used in the study

3.2. Environmental and Remote Sensing Data

Index / Dataset	Purpose	Platform
NDVI	Vegetation stress; crop and ecosystem health monitoring	Google Earth Engine
NDWI	Surface water presence and seasonal dynamics	Google Earth Engine
SPI / SPEI / SMA	Meteorological and agricultural drought quantification	Google Earth Engine
LST (Sentinel-2 / MODIS)	Heat stress; groundwater depletion correlation	Google Earth Engine
Sentinel-2 L2A	Multispectral imagery: water turbidity, land use, salinity indicators	Copernicus Browser
Climate Historical Data	Rainfall variability, temperature trends	World Bank Climate Portal

Table 2: Environmental and remote sensing datasets

3.3. Socioeconomic Data

Dataset	Description	Source
INS Population Data	Population density, rural/urban classification, household statistics	Tunisia National Open Data Portal
UNICEF Tunisia Extracts	Water access rates, sanitation coverage, vulnerability indicators	UNICEF Tunisia
Agricultural Land Use	Irrigated vs. rain-fed areas, crop types, farming community locations	FAO / INS
Water Service Interruption Logs	Frequency and duration of supply cuts by zone	SONEDE (national utility)
Health Outcome Statistics	Waterborne disease incidence linked to poor water quality	Ministry of Health / WHO

Table 3: Socioeconomic data sources

4. Data Preprocessing

4.1. Spatial Data Harmonisation

All vector and raster datasets were projected to a common coordinate reference system (**WGS84 / UTM Zone 32N**) to ensure geometric consistency. Administrative boundary layers were clipped to the Sfax Governorate extent, and topology errors in OpenStreetMap extracts were corrected using QGIS validation tools. Water infrastructure points were deduplicated and tagged with service type (network connection, standpipe, well, tanker supply).

4.2. Remote Sensing Preprocessing in Google Earth Engine

Sentinel-2 Level-2A imagery was filtered by cloud cover ($< 20\%$) and composited using median pixel values over the 2020–2025 period to minimise noise. The following indices were then computed per pixel:

- NDVI $= (B_{08} - B_{04})/(B_{08} + B_{04})$
- NDWI $= (B_{03} - B_{08})/(B_{03} + B_{08})$

- LST derived from Band 11 (SWIR) calibrated against MODIS land surface temperature products

SPI and SPEI were computed from ERA5 monthly precipitation and temperature reanalysis data, aggregated over 3-, 6-, and 12-month windows. Standardised anomalies were classified into drought severity categories following the McKee (1993) scale.

4.3. Socioeconomic Data Integration

Tabular socioeconomic indicators (population density, water access rates, agricultural land area) were joined to administrative unit polygons (delegation level) using unique identifiers. Missing values for rural sub-districts were imputed using spatial interpolation (inverse distance weighting) validated against UNICEF district-level extracts.

4.4. DRASTIC Parameter Rasterisation

The seven DRASTIC parameters were sourced from the following layers and rasterised at 100 m resolution:

Parameter	Name	Data Source
D	Depth to water	Hydrogeological borehole logs
R	Recharge rate	GRACE / ERA5 groundwater recharge estimates
A	Aquifer media	Geological survey maps
S	Soil media	FAO Harmonised World Soil Database
T	Topography	SRTM 30 m DEM
I	Vadose zone impact	Lithological profiles
C	Hydraulic conductivity	Pump test records

Table 4: DRASTIC model parameters and data sources

5. Spatial Analysis Steps

5.1. Step 1 — Water Accessibility Mapping

Water infrastructure points (from OSM and national data) were kernel-density interpolated to produce continuous service coverage surfaces. Access scores per settlement were computed by overlaying coverage surfaces with population distribution using the formula:

$$\text{Access Score}_i = \frac{\text{Population with network access}_i}{\text{Total population}_i} \times 100$$

Urban–rural disparities were visualised by choropleth maps at delegation level, with supply interruption frequency overlaid as a categorical layer.

5.2. Step 2 — Water Quality Assessment

Salinity concentration contours were generated from borehole water quality records using ordinary kriging interpolation. The resulting salinity surface was classified into five suitability classes (WHO standards for drinking and FAO standards for irrigation). Contamination risk was mapped by overlaying industrial facility buffers with groundwater flow direction rasters derived from the DEM.

5.3. Step 3 — DRASTIC Groundwater Vulnerability Index

Each parameter was rated and weighted according to the standard DRASTIC protocol:

$$\text{DRASTIC Index} = D_R D_W + R_R R_W + A_R A_W + S_R S_W + T_R T_W + I_R I_W + C_R C_W$$

where R denotes the rating and W the weight for each parameter. The resulting index was reclassified into four vulnerability zones: *Low*, *Moderate*, *High*, and *Very High*.

5.4. Step 4 — Climate Stress Analysis

Climate Stress Index (CSI) integrates three normalised indicators:

$$\text{CSI} = \frac{1}{3}\left(T_{\text{norm}} + \text{RV}_{\text{norm}} + \text{DF}_{\text{norm}}\right)$$

where T_{norm} = temperature increase, RV_{norm} = rainfall variability, DF_{norm} = drought frequency.

SPI/SPEI/SMA rasters were averaged across the 2015–2025 decade to produce a chronic drought exposure surface. LST anomalies above 40°C were classified as extreme heat zones. NDVI trend slopes (linear regression per pixel) identified areas of progressive vegetation degradation.

5.5. Step 5 — Water Scarcity Index (WSI)

$$\text{WSI} = f\,(\text{Access coverage},\ \text{Groundwater depletion},\ \text{Salinity},\ \text{Interruption frequency})$$

Each component was normalised to $[0, 1]$ and averaged using equal weights in the baseline model (sensitivity analysis tested alternative weightings).

5.6. Step 6 — Climate–Water Vulnerability Index (CWVI)

The composite vulnerability index was derived as:

$$\text{CWVI} = \text{CSI} \times \text{WSI}$$

This multiplicative approach ensures that zones scoring high on both climate stress and water scarcity receive the highest vulnerability scores, while areas with only one stressor are treated with intermediate priority.

5.7. Step 7 — Food Security Risk Overlay

An additional Food Risk Index (FRI) was computed to integrate SDG 2 considerations:

$$\text{FRI} = \text{WSI} \times \text{CSI} \times \text{Agri\_Dep} \times \text{GW\_Vuln}$$

where **Agri_Dep** is the proportion of the local economy dependent on irrigated agriculture and **GW_Vuln** is derived from the DRASTIC index. The FRI was overlaid with NDVI stress rasters to validate agricultural impact areas.

5.8. Step 8 — Hotspot Identification and Prioritisation

CWVI and FRI rasters were combined with population density to weight vulnerability by social exposure:

$$\text{Priority Score} = \text{CWVI} \times \log(\text{Pop. Density} + 1)$$

Hotspots were defined as cells in the top quintile of the Priority Score distribution. These were aggregated to delegation level to enable policy communication and budget targeting.

6. Outputs

6.1. Final Water Scarcity Vulnerability Map

The primary cartographic output is a multi-layer vulnerability map of Sfax Governorate rendered in QGIS and published via a Streamlit dashboard. The map comprises the following layers:

- **Layer 1 — Water Access Coverage:** Network connection rates by settlement (choropleth, delegation level), overlaid with infrastructure points classified by service type.
- **Layer 2 — Water Quality Zones:** Salinity and contamination suitability rasters (5 classes for drinking; 4 classes for irrigation).
- **Layer 3 — Groundwater Vulnerability (DRASTIC):** Continuous DRASTIC index surface reclassified into Low/Moderate/High/Very High categories.
- **Layer 4 — Climate Stress (CSI):** Drought exposure, LST anomaly, and NDVI degradation trend composite.
- **Layer 5 — Composite CWVI:** Final vulnerability surface showing convergence zones of climate and water stress.
- **Layer 6 — Food Security Risk (FRI):** Agricultural vulnerability hotspots for SDG 2 policy targeting.
- **Layer 7 — Priority Intervention Zones:** Top-quintile hotspots weighted by population exposure, symbolised with a ranked colour scheme for immediate policy use.

6.2. Interactive Dashboard

The Streamlit dashboard integrates the maps with a decision-support interface built on the GEE → Python → Streamlit pipeline. Key features include:

Gamification / Impact Estimator:

"If you invest in Zone X →

- **X people** gain safe water access
- **X hectares** of farmland secured
- **X farming households** protected
- **X people** protected from food insecurity"

7. Recommendations

7.1. Priority Intervention Zones

Based on the spatial analysis results, the following priority zones are identified for targeted policy action:

Zone Type	Priority Tier	Recommended Intervention
Rural inland communities (Sfax–Agareb basin)	**Tier 1**	Emergency safe water supply extension; borehole rehabilitation with quality treatment units
lightgray Coastal peri-urban fringe	**Tier 2**	Seawater intrusion monitoring network; industrial effluent control; aquifer recharge schemes
Agricultural hotspot zones (high FRI)	**Tier 2**	Drip irrigation rollout; groundwater allocation caps; drought-resistant crop promotion
lightgray Dispersed rural settlements ($>$ 2 km from network)	**Tier 1**	Community rainwater harvesting; mobile water treatment units; solar-powered pumping
High DRASTIC vulnerability aquifer zones	**Tier 2**	Pollution source buffer zones; contamination early-warning sensors; protection perimeter legislation

Table 5: Priority intervention zones and recommended actions

7.2. Infrastructure Investment Priorities

R1. Extend piped water networks to the 15% of rural delegations with network coverage below 50%, prioritising zones in Tier 1 CWVI clusters.

R2. Install decentralised treatment units in coastal aquifer zones where salinity exceeds WHO drinking water limits ($> 1{,}000\ \mu$S/cm).

R3. Rehabilitate and monitor boreholes in the Sfax–Agareb basin, fitting all extraction points with flow meters and IoT water quality sensors.

R4. **Establish aquifer protection perimeters** around the top-five most vulnerable DRASTIC zones, with enforceable land-use restrictions.

R5. **Invest in drought-resilient irrigation** infrastructure in agricultural hotspot zones, targeting farms exceeding 50% groundwater dependency.

7.3. Proposed SDG 6 Monitoring Framework

A spatially-explicit monitoring framework is proposed, structured around three tiers aligned with SDG 6 global indicators:

Tier A — Access Monitoring (SDG 6.1 & 6.2)

- Annual re-mapping of water access rates at delegation level
- Biannual field surveys in priority zones to validate remote sensing outputs
- Real-time supply interruption tracking via SONEDE operational data

Tier B — Quality & Aquifer Monitoring (SDG 6.3 & 6.6)

- Quarterly borehole water quality sampling (salinity, nitrates, biological contamination)
- Annual NDWI-based surface water change detection in GEE
- Triennial groundwater level surveys to track depletion rates

Tier C — Climate Resilience Monitoring (SDG 6.5 & SDG 13)

- Monthly drought index (SPI/SPEI) updates via automated GEE scripts
- Annual CWVI recalculation to track vulnerability trends
- Scenario-based modelling updates every 3 years using new climate projections (CMIP6)

8. SDG Linkages

8.1. SDG 6 — Clean Water and Sanitation

SDG 6: Clean Water & Sanitation

How this project operationalises SDG 6:

- **Target 6.1** (safe drinking water): Water access maps identify the proportion of the population lacking safely managed services and locate the most underserved settlements.
- **Target 6.3** (water quality): Salinity and contamination maps quantify the share of water bodies meeting safe quality standards and guide treatment investment priorities.
- **Target 6.4** (water use efficiency): Agricultural water dependency mapping and FRI scores support allocation policies that increase efficiency while protecting food production.
- **Target 6.5** (integrated water management): The CWVI and DRASTIC layers provide the evidence base for integrated catchment-level governance, enabling cross-sectoral coordination.
- **Target 6.6** (water-related ecosystems): NDWI and LST monitoring tracks surface water body health, wetland coverage, and the impacts of extraction on ecosystem integrity.

The project directly contributes to **SDG 6 monitoring** by producing spatially explicit baseline maps that can be updated annually, enabling Tunisia to track progress against national water security targets with geographically disaggregated evidence. By linking water access data to poverty and health indicators, the analysis also provides the factual foundation for equitable resource allocation decisions.

8.2. SDG 13 — Climate Action

SDG 13: Climate Action

How this project operationalises SDG 13:

- **Target 13.1** (resilience and adaptive capacity): CWVI hotspot maps allow authorities to identify where climate-induced water stress is highest, enabling proactive resilience investments before crises occur.
- **Target 13.2** (climate in national policies): The project provides the spatial intelligence needed to integrate climate risk into Tunisia's national water and agricultural policies.
- **Target 13.3** (climate awareness and capacity): The GEE → Python → Streamlit pipeline makes satellite-derived climate indicators accessible to non-specialist decision makers through an interactive dashboard.

Climate change is not treated as background context in this project but as a *measurable, spatially explicit driver* of water insecurity. By connecting climate signals (drought indices, LST anomalies, NDVI degradation) to hydrological outcomes (groundwater depletion, service disruption) and social vulnerability (rural isolation, agricultural dependency), the project makes SDG 13 a quantified component of the analysis. The early-warning-style drought and heat stress maps are directly deployable in climate adaptation planning at the governorate and national levels.

8.3. Supporting SDGs

Beyond the two primary frameworks, the project contributes evidence to four additional goals:

SDG	Contribution
SDG 2 — Zero Hunger	FRI maps identify agricultural zones at risk; salinity and NDVI analysis directly informs food security planning for farming communities dependent on groundwater.
SDG 3 — Good Health	Contamination and salinity maps identify populations exposed to waterborne health risks, enabling targeted public health interventions.
SDG 11 — Sustainable Cities	Urban–rural access disparity maps reveal infrastructure gaps in peri-urban zones, informing inclusive urban water planning.
SDG 15 — Life on Land	DRASTIC vulnerability and NDWI layers protect aquifer recharge zones and wetlands from degradation through evidence-based land-use management.

Table 6: Additional SDG contributions

9. Conclusion

This project demonstrates how geospatial engineering can transform scattered environmental, hydrological, climatic, and socioeconomic data into coherent spatial intelligence for sustainable development. The Sfax region, characterised by deep water access inequalities, degrading coastal aquifers, climate-driven inland drought, and significant agricultural water dependency, represents a high-priority case for this type of integrated analysis.

The Water Scarcity Vulnerability Map produced by this project, grounded in the CWVI framework and validated by DRASTIC modelling, provides decision makers with a clear, evidence-based picture of where intervention is most urgently needed and where it will deliver the greatest social return. The proposed SDG 6 monitoring framework ensures these findings are not a one-time snapshot but the foundation of a continuously updated spatial observatory.

By linking water access to food security (SDG 2), public health (SDG 3), and climate resilience (SDG 13), the project confirms the core thesis of the geomatics approach to sustainable development: *maps are not just descriptions of a problem — they are instruments of solution.*

"From maps to action: this project delivers spatial intelligence for policy, enabling authorities to prioritise investments, protect vulnerable aquifers, improve rural access, and design climate-resilient water strategies."

Key References and Data Sources

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